General Physics I — PHYS 2500-02

Fall 2026

CSUSB Land Acknowledgement: We recognize that California State University, San Bernardino sits on the territory and ancestral land of the San Manuel Band of Mission Indians (Yuhaaviatam). We recognize that every member of the California State University, San Bernardino community has benefitted and continues to benefit from the use and occupation of this land since the institution's founding in 1965. Consistent with our values of community and diversity, we have a responsibility to acknowledge and make visible the university's relationship to Native peoples. By offering this Land Acknowledgement, we affirm Indigenous sovereignty and will work to hold California State University, San Bernardino more accountable to the needs of American Indian and Indigenous peoples.

Instructor Information

Instructor: Kevin Collins
Office location:
Telephone:
Email:
Office hours: Tuesday 00:00MM – 00:00MM, Friday 00:00MM – 00:00MM
Class Days/Time: Tuesday, Thursday, Friday 10:30AM – 11:45AM
Classroom:
Class Web Page: Canvas

Course Description

First course of a two-course sequence in introductory calculus-based physics for scientists and engineers. This sequence is intended for students with a strong background in mathematics and the sciences. Topics include mechanics and oscillations. Satisfies GE B1/5A.

- **Prerequisite**: MATH 2210
- **Corequisite**: MATH 2220

Lab 2500L
Most of you will take PHYS 2500L concurrently with this course. It is a separate class with its own registration, instructor, and grade — your lab work does not affect your grade in this course, and this course does not affect your lab grade. Check with your lab instructor about that course's requirements.
I am not the instructor of record for the lab. I'm glad to hear feedback or concerns about lab sections, but I have limited control over them. If there's an issue you aren't comfortable raising with your lab instructor, the department chair is usually the most productive person to talk to.

Required Text and Course Materials

Physics for Scientists and Engineers: A Strategic Approach with Modern Physics, 5th edition, by Randall D Knight – ISBN: 978-0137302260 or 978-1292438221
You can get the book new or used, physical or digital, from the Coyote Bookstore, Pearson, Amazon, eBay, or anywhere else you like.
Older editions will generally work, but chapter and section numbers differ. If you use one, it is your responsibility to match the reading assignments and homework problems to your edition.

Mastering Physics is not used in this course, and you do not need to buy an access code.

Canvas and Gradescope
All course materials are posted on Canvas, and homework is submitted through Gradescope. For technical support with canvas visit the Technology Support Center located on the 1st floor of the Pfau Library Wedge, PL1108. They can also be reached by calling 909-537-7677.

Calculator
You need a scientific calculator — one with log, sine, cosine, square root, and similar functions. During class, use whatever is convenient, including your phone. On unit assessments and the final, you may use only a scientific calculator: no phones, no graphing calculators, and no device that can store data or connect to a network.

Grading

Assignment weight:

Participation	10%
Homework	10%
Unit assessment 1	20%
Unit assessment 2	20%
Unit assessment 3	20%
Final	20%

Letter grade breakdown:

A	93–100	B-	80–82.9	D+	67–69.9
A-	90–92.9	C+	77–79.9	D	63–66.9
B+	87–89.9	C	73–76.9	D-	60–62.9
B	83–86.9	C-	70–72.9	F	59.9 or less

TL;DR

- Tuesday is the problem-solving session. Thursday and Friday are lectures.
- Homework is assigned each Tuesday and is due at 11:59 p.m. on the day the next homework is assigned — usually the following Tuesday. See the schedule.
- Homework is submitted through Gradescope.
- Homework is graded on effort and completeness, not on getting the right answer.
- No late homework, but your lowest homework score is dropped automatically.
- Problem-solving sessions are graded on participation, not correctness. They normally cannot be made up, but your lowest score is dropped automatically — no explanation needed. See Course Structure for absences covered by university policy.
- Attendance is not tracked. Your participation grade is the only place showing up affects your grade.
- AI is allowed as a tutor, not as a substitute. Use it to get unstuck; don't use it to produce work you didn't do.
- You may work with classmates on homework, but submit your own work — and be part of doing it.
- Unit assessments and the final are in class, closed book, scientific calculator only. Together they are 80% of your grade.
- Mastering Physics is not used. Don't buy an access code.

This is a summary. The full policies are below. Read them.

Learning Outcomes

By the end of this course, you should be able to:

- **Represent a physical situation.** Sketch the system, identify the objects, interactions, and relevant physical features, define a coordinate system, and state the assumptions and idealizations being made.
- **Supply what isn't given.** Identify information that is needed but not stated, and supply it from physical knowledge, standard values, or reasonable estimation.
- **Solve problems mathematically.** Construct and carry out a solution using vector methods and calculus, applying kinematics, Newton's laws, and the conservation of energy, momentum, and angular momentum.
- **Check your own work.** Evaluate a result for physical reasonableness — units, order of magnitude, limiting and special cases, symmetry — and revise the solution when it fails those checks.
- **Explain your reasoning.** Say why a given principle applies, why a term is negligible or zero, and what an assumption buys and what it costs.
- **Communicate a solution.** Present your work in writing and orally so that another student can follow it, and move between verbal, graphical, diagrammatic, and algebraic representations of the same situation.
- **Work productively in a group.** Share your reasoning aloud, respond to others' approaches, and help a peer locate the source of a difficulty rather than supplying the answer.

My Commitments to You

Physics is hard. It's also very interesting. When you finally see how the pieces fit together and how they describe everyday phenomena around you, it's genuinely satisfying. My goal is to get you there. To that end, I've structured this course to build deliberately. The weekly rhythm of lecture, group problem solving, and homework is designed to give you repeated chances to use what you've learned rather than meet it once and move on, and I'll keep adjusting that structure as the term goes based on what this class actually needs. I believe everyone here is capable of succeeding. Physics rewards persistence more than it rewards natural talent, and my part of the deal is to make sure the effort you put in actually goes somewhere.

Course Structure

This course meets three times a week. Tuesdays are problem-solving sessions. Thursdays and Fridays are lectures.

Problem-Solving Session (Tuesday)

You'll work in small groups on problems from the previous week's material while I circulate to help. These problems are richer than what you'll see on assessments — often without a tidy list of givens, sometimes requiring you to supply information yourself. We'll usually get through about three in a session.

These problems are related to the homework but aren't the homework itself. They cover the same physics in different settings, so attempting the homework first is what makes the session most useful.

I may assign you to a group or move you between groups during the term. All group members are expected to participate. After groups have worked a problem, I'll either present the solution or call on a group to present it at the board — you should expect to be asked to present at some point during the term.

This is meant to be a place to make mistakes safely. Getting a problem wrong here costs you nothing.

Participation credit is based on taking part in these sessions: working the problems with your group, engaging with your groupmates' reasoning, and presenting at the board when your group is called on. It is not based on getting the right answer. Being disruptive or disengaged during any class meeting can affect this grade. Because the credit is for participating in the session itself, a missed session normally can't be made up, and your lowest score is dropped automatically — no documentation or explanation needed. If you have an absence covered by CSUSB's Class Attendance Policy (FAM 818.9) — illness, a death or serious illness in your immediate family, religious observance, a university-sanctioned activity, or jury duty or military service — email me as soon as you know, and I'll assign you alternative work to make up the session.

Lecture (Thursday, Friday)

New material is presented, with short targeted examples showing each concept in action. Lectures also include clicker questions — multiple-choice questions you answer in class by holding up a folded card marked A, B, C, or D — so both you and I can see how well you understand the material. These aren't graded. They exist to find confusion early, and a wrong answer that surfaces a misunderstanding is more useful than a right one. Because Tuesdays are reserved for problem solving, lectures move at a steady pace. Reading the assigned sections before class makes a real difference, and coming in with questions makes an even bigger one.

Homework

Homework is assigned each Tuesday and covers that week's lectures. It is due at 11:59 p.m. on the day the next homework is assigned — usually the following Tuesday, though a few weeks differ. Homework is submitted through Gradescope; see the schedule for dates.

Mastering Physics is not used in this course.

Homework is graded on effort and completeness, not on correct answers. A serious attempt at a problem you got wrong earns full credit; a blank page doesn't. I know this isn't your only class, so I've aimed for enough problems to give you real practice without the load being unreasonable.

No late homework is accepted, and there are no makeups. Your lowest homework score is dropped automatically. That covers illness, emergencies, and the occasional bad week — you don't need to ask, and you don't need to explain.

You're welcome to work with classmates. But submit your own write-up and be part of producing it — copying someone else's solutions, or having a chatbot produce them, isn't collaboration and doesn't do anything for you.

Using a chatbot as a tutor is fine and can be genuinely useful: ask it to explain a concept, check your reasoning, or walk you through something the textbook didn't make clear. What doesn't work is handing it a problem before you've attempted it yourself. Struggling with a problem for ten minutes before getting help is where most of the learning happens. You need to know this material yourself. Homework is 10% of your grade; assessments and the final are 80%, and you'll do those with no outside help — your grade will reflect what you actually understand.

Unit Assessments

There are three unit assessments, each covering the unit just completed (see schedule for units). They're given in class during the regular class period, closed book, with a scientific calculator only — no phones, no graphing calculators, no devices that store data or connect to a network.

Assessments test the core ideas of the unit rather than the harder problems we work on during the Tuesday sessions. Assessment dates are on the schedule, and I'll go over the format in class beforehand.

Missing a unit assessment: If you know in advance that you'll miss an assessment, email me as soon as possible — ideally before the assessment. If something happens you couldn't plan for, contact me as soon as you're able. Absences covered by FAM 818.9 will be accommodated; for anything else, I'll work out an arrangement with you individually. Either way, tell me early — the sooner I know, the more options there are.

Final Exam

The final is cumulative, with particular emphasis on oscillations. The same rules apply as for unit assessments, and I'll go over the format in class before the end of the term.

Tips for Success

Physics builds on itself. You'll be expected to use material from the start of the term to solve problems with material from the end of it, so falling behind compounds and staying current pays off more than it does in most classes.

- **Do the homework** — and attempt it before Tuesday's session.
- **Come to the problem-solving sessions.** They're the most direct practice you'll get for what assessments look like.
- **Work with your classmates.** Explaining something to someone else is one of the fastest ways to find out whether you actually understand it.
- **Read ahead.** Follow the schedule, read the section before lecture, and come with questions.
- **Come to office hours.** Bring a problem you're stuck on. You don't need to have it figured out first — that's the point.

Course Policies

Classroom environment

This class only works if people are willing to say they don't understand something. Nobody in this room gets to make anyone feel bad for being confused, asking a basic question, or getting something wrong — and that includes me. Respect, patience, and good faith toward your classmates are expected at all times.

Attendance and participation

Attendance isn't tracked. Many of you have responsibilities outside this class, and if you're ill, the right thing to do is stay home rather than pass it around. How to spend your time is your call. The exception is the problem-solving sessions, which carry a participation grade that normally can't be made up. That grade is 10% of your course total, so missing sessions won't sink you on its own — but there's a clear connection between showing up, working with other people, and understanding the material. I'd encourage you to come, and I'll do my part to make the time worth spending.

Technology

Use technology for class-related work whenever it helps — including your phone as a calculator. Phones and laptops used for anything else are a distraction to the people around you, and if that's what's happening, you'll be asked to put it away.

Working with others

Collaboration is expected in the problem-solving sessions and encouraged on homework. The work you turn in has to be your own: your own write-up, produced with your own participation. Unit assessments and the final are entirely individual.

Changes to this syllabus

Everything here is tentative and subject to change, including the schedule, due dates, assessment dates, and course policies. I'll adjust as the term goes based on what the class needs. If anything changes, I'll announce it in class and post it on Canvas — you won't be held to a change you weren't told about.

If something isn't working

Tell me. During class, at office hours, or by email. I'd rather hear about a problem in week four than read about it on an evaluation in December.

University Policies

Plagiarism and Cheating

Students are expected to be familiar with the University's Policy on cheating and Plagiarism. Please review this at https://catalog.csusb.edu/academic-regulations/ under the heading "Plagiarism and Cheating:

"Plagiarism and cheating are violations of the Standards for Student Conduct (Title 5, §41301, California Code of Regulations). Definitions and procedures for addressing cheating and plagiarism are found in the CSUSB Faculty Senate Policies and Procedures Concerning Academic Dishonesty (FAM 803.5) and the California State University Student Conduct Process (Executive Order 1098) and may be addressed by both the instructor and the Student Conduct Administrator… As described in FAM 803.5, plagiarism is the act of presenting the ideas and writing of another as one's own. Cheating is the act of obtaining or attempting to obtain credit for academic work through the use of any dishonest, deceptive, or fraudulent means."

Instances of academic dishonesty will not be tolerated. Cheating on exams or plagiarism will result in a failing grade and sanctions by the University. For this class, all assignments are to be completed by the individual student unless otherwise specified.

Dropping and Adding
You are responsible for understanding the policies and procedures about add/drops, academic renewal, etc. found at https://catalog.csusb.edu/academic-regulations/

Support for Students with Disabilities

If you are seeking classroom accommodations under the ADA, you are required to register with the Office of Services to Students with Disabilities (SSD). If you are taking classes at the San Bernardino Campus (SBC), SSD is located in UH-183. At the Palm Desert Campus (PDC), SSD is located in RG 203A. You can telephone the SBC-SSD at (909) 537-5238, or the PDC-SSD at (760) 341-2883, ext. 78117, or email them at ssd@csusb.edu. To receive academic accommodations for this class, please obtain the Faculty Notification Letter from SSD.

Students should inform their instructors about the type of assistance that may be required in the event of any emergency, and also to contact the SSD office regarding special needs for evacuation.

Statement on Inclusion, Diversity, and Equity

The California State University does not discriminate on the basis of age, genetic information, marital status, medical condition, nationality, race or ethnicity (including color and ancestry), religion (or religious creed), and veteran or military status – as these terms are defined in CSU Executive Order 1097– in its programs and activities, including admission and access. Federal and state laws, including Title VI of the Civil Rights Act of 1964 and the California Equity in Higher Education Act, prohibit such discrimination.

The California State University does not discriminate on the basis of sex, gender, or sexual orientation in its education programs or activities. Title IX of the Education Amendments of 1972, and certain other federal and state laws, 1 prohibit discrimination on the basis of sex, gender, or sexual orientation in employment, as well as all education programs and activities operated by the University (both on and off campus), and protect all people regardless of their gender or gender identity from sex discrimination, which includes sexual harassment and sexual violence.

Course Schedule

(Note: subject to change with fair notice.)

- "HW#" means Homework # is assigned that day, and the previous homework is due. Homework is always due by 11:59 p.m. on the day the next homework is assigned.
- Weeks 1, 12, and 13 have no problem-solving session. In week 15, the session moves to Thursday.

Date	Tuesday	Thursday	Friday
Week 1 (08/24 – 08/28) **Vectors, units, physics problem solving**	3.1, 3.2 HW1	3.3, 3.4	1.8, 1.7
Week 2 (08/31 – 09/04) **Motion and kinematics**	Problem-solving session HW2	1.1 – 1.3	1.4 – 1.6
Week 3 (09/07 – 09/11) **Motion and kinematics**	Problem-solving session HW3	2.1 – 2.3	2.4 – 2.6
Week 4 (09/14 – 09/18) **Motion and kinematics**	Problem-solving session HW4	4.1, 4.2	4.3 – 4.6
Week 5 (09/21 – 09/25) **Force, motion, and Newton's laws**	Problem-solving session HW5	5.1 – 5.3	5.4 – 5.7
Week 6 (09/28 – 10/02) **Force, motion, and Newton's laws**	**Unit assessment 1 - Motion and kinematics**	6.1 – 6.3 HW6	6.4, 6.6
Week 7 (10/05 – 10/09) **Force, motion, and Newton's laws**	Problem-solving session HW7	8.1 – 8.3	8.4, 8.5
Week 8 (10/12 – 10/16) **Force, motion, and Newton's laws**	Problem-solving session HW8	7.1 – 7.3	7.4 – 7.5
Week 9 (10/19 – 10/23) **Conservation laws (energy and momentum)**	Problem-solving session HW9	9.1 – 9.3	9.4 – 9.6
Week 10 (10/26 – 10/30) **Conservation laws (energy and momentum)**	**Unit assessment 2 - Force, motion, and Newton's laws**	10.1 – 10.3 HW10	10.4, 10.6 – 10.8
Week 11 (11/02 – 11/06) **Conservation laws (energy and momentum)**	Problem-solving session HW11	11.1, 11.2	11.3 – 11.5
Week 12 (11/09 – 11/13) **Applications of Newtonian mechanics**	12.1 – 12.4 HW12	12.5 – 12.7	12.10, 12.11
Week 13 (11/16 – 11/20) **Applications of Newtonian mechanics**	**Unit assessment 3 - Conservation laws**	15.1 – 15.3 HW13 (due 12/3)	15.4 – 15.6
Week 14 (11/23 – 11/27)	NO CLASS NO HW	NO CLASS	NO CLASS

Date	Tuesday	Thursday	Friday
Week 15 (11/30 – 12/04) **Applications of Newtonian mechanics**	15.7, 15.8	Problem-solving session	NO CLASS
Week 16 (12/07 – 12/11)	**FINAL** **10:30AM – 12:30PM**		